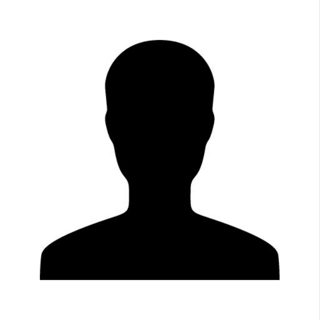
Building a Quantum Circuit

Using IBM Quantum Composer

Presenters



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Using the Qubit

The Qubit can be described as two mathematical vectors.

What we have described here is the abstract notion of a qubit.

IBM quantum computers use ***superconducting transmon qubit***, which is made from superconducting materials such as niobium and aluminum, patterned on a silicon substrate.

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

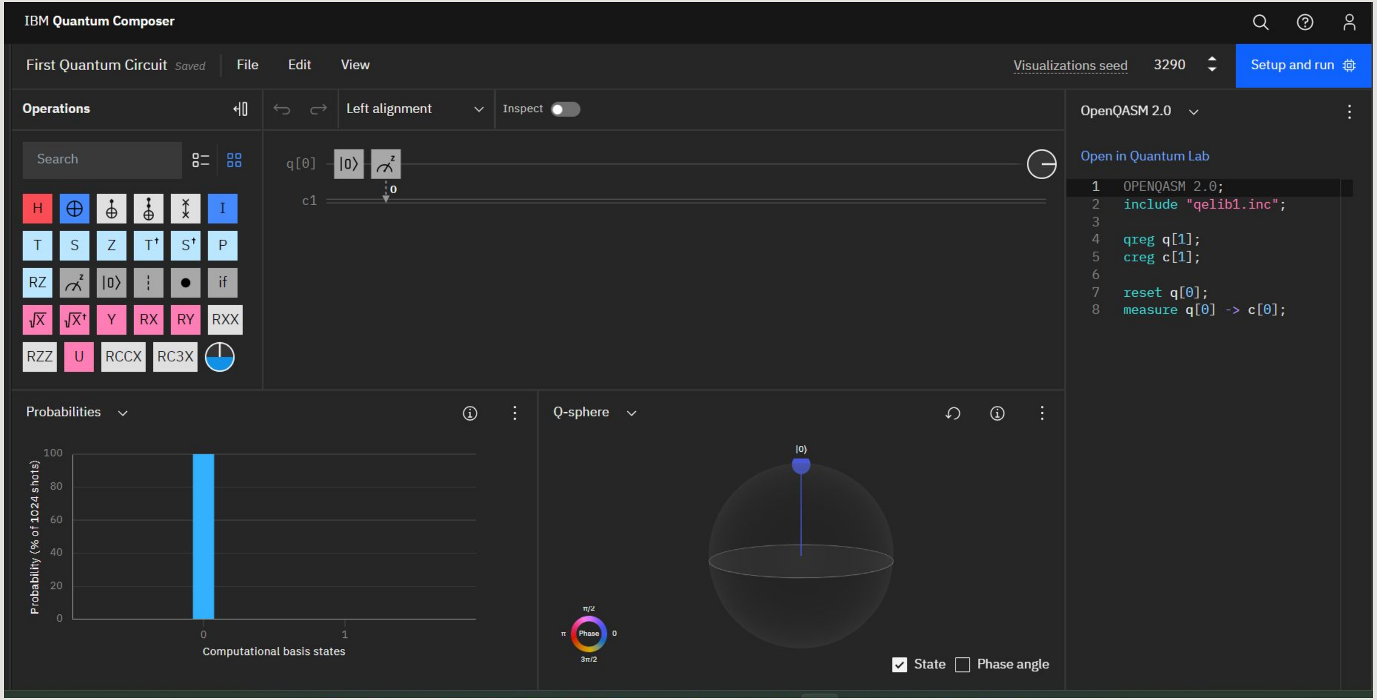
Using the Qubit

For a superconducting qubit to behave as the abstract notion of the qubit, we must have the device at drastically low temperatures such as 15 millikelvin. Once this is done, the qubit is said to be at ground state, or 0.

$$|0\rangle$$

The reset operation returns a qubit to ground state irrespective of its state before the operation was applied.

Using the Qubit



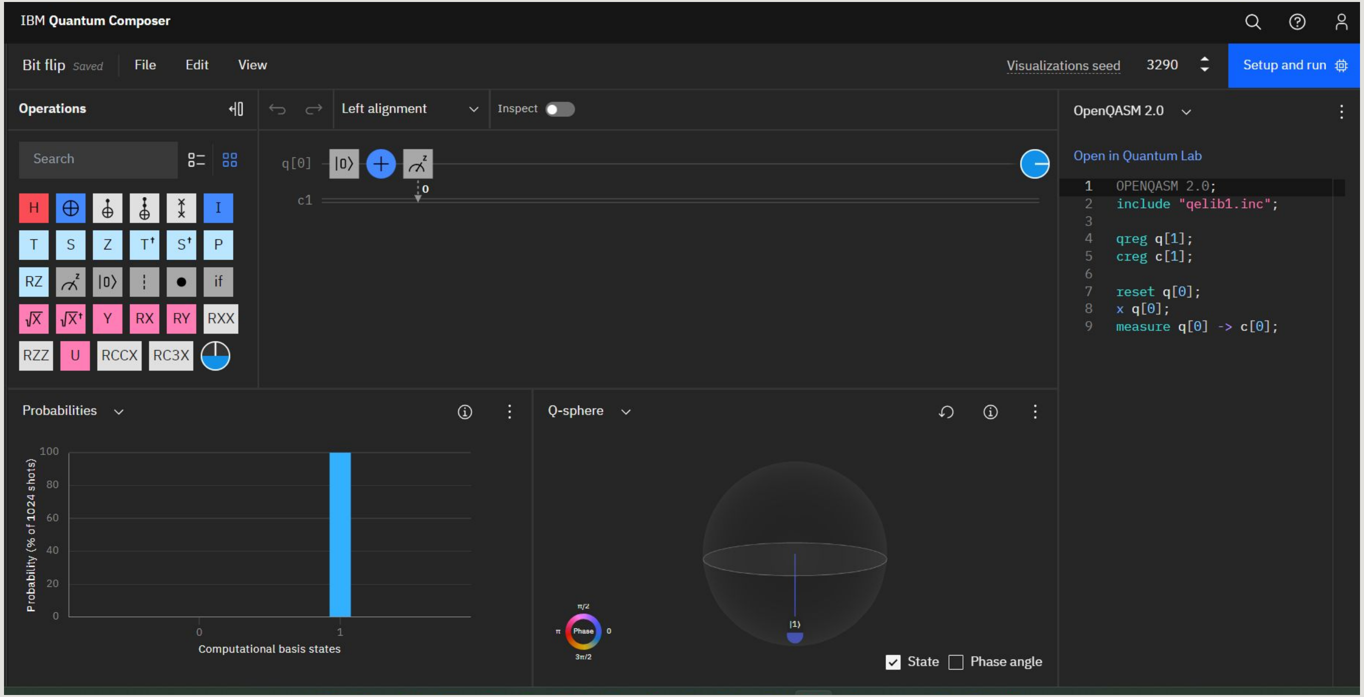
Using the Qubit

We now need to be able to put the qubit in other states. To do this, we require the concept of a ***quantum gate.*** A quantum gate is a 2x2 matrix. Quantum gates are a way to describe a quantum state evolution. The matrix transforms an initial state to a final state. This is simply a matrix transformation.

NOT Gate

Flips the $|\mathbf{0}\rangle$ state to $|\mathbf{1}\rangle$, and vice versa.

Using the Qubit



References

https://quantum-computing.ibm.com/composer/files/2a126cbf533fbbf8ce60e84b2616a0043782015916bc85cae16825a7327bb39d

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